

# Regression Utility of a 10,240-Byte Tabular Generator

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**Abstract**—This paper measures how much of a real-trained regressor’s improvement over a null is recovered by an auditor trained on synthetic rows. The generator is evaluated against a 10,240-byte artifact budget. It is a compact residual, not a GAN, a diffusion model, or a transformer: at most 12 screened inputs, a width-16 tanh layer, a refit readout, and 2,048 AdamW steps at learning rate 0.002, with no embedding. The configuration id is `features12_steps2048`. The panel is 100 prepared PMLB regression lineages, including simulated physics and simulated dynamics, with a grouped 80/20 cut of the official training rows, seed 1729, fit seed 11, and synthetic size  $4n$ . On 97 informative lineages the CatBoost median retention is 0.940 (95% lineage-bootstrap interval 0.914 to 0.976). Holm-adjusted Wilcoxon tests separate this generator from a Gaussian copula, a Chow–Liu tree, and independent marginals. Against the copula the record is 86/0/11 wins, ties, and losses. The linear auditor does not separate the two (49/0/43). That interval contains zero, and the result is not an equivalence test. Unpooled blocks compare CTGAN and TVAE on 21 lineages and Forest-Flow on 6 lineages (1/0/5; 115,202,253 to 1,085,155,062 bytes). 98/100 artifacts are inside the cap, and two are over it. Official tests were not opened. TabSyn is measured on a retained validation subset, the 100-lineage TabDDPM publication is not measured, and the full Forest-Flow publication is not measured. Fit-seed variance is not measured. Release scores are null. The distance ledger is not differential privacy or HIPAA de-identification.

**Index Terms**—Synthetic tabular data, retention, artifact bytes, paired tests.

## I. INTRODUCTION

A generator paper often reports the metric that also selected the configuration. Those native scores live on different scales: predictive efficacy for CTGAN and TVAE [1], held-out log density for a Gaussian copula [2], a forest density for ARF [3], and a diffusion or CART objective for other families [4], [5]. This paper separates selection from the common measurement. After each method has been selected on its own objective, the shared columns are retention and the bytes charged to the artifact.

The decision those columns can support is narrow. It is whether a network small enough to ship under a 10,240-byte cap still preserves downstream regression utility on held-out validation rows. It is not a declaration that one generator is best, and it is not a certification score. Master Fitness v2 (MFS-v2), a gated geometric score, is defined in Section III-D. It is null on every method here, because the six components and the hard-gate map are not all present.

The contributions are:

- 1) The displayed residual and the generative procedure that samples a row from that artifact (Section III, Fig. 1, (1)). That procedure is not the production default and it does not use the 4,168-dimensional router.
- 2) Retention and charged bytes on the validation panel, with lineage-bootstrap intervals, and the algebra of a retention value (Fig. 2, Fig. 3).
- 3) Paired tests inside unpooled blocks: the density block, the neural block, and the 6-lineage Forest-Flow confirmation, including the linear non-separation and the Forest-Flow lead (Fig. 4).

*a) Scope.*: TabSyn is measured on a retained validation subset. The 100-lineage TabDDPM publication is not measured. The full Forest-Flow publication is not measured. The pre-specified privacy-attack panel is not measured. Fit-seed variance

is not measured. The full ablation grid is not measured. The fits use one fit seed, 11, and coresets of 72 to 800 rows. Official tests stay sealed. MFS-v2 is null. PTF-v1, the release KPI whose hard gates include joint fidelity, membership and attribute attacks, copy checks, and feature-importance agreement on a sealed holdout, is null. A release at the 10,240-byte tier is a separate claim from the byte cap, and it is not made here. BeyondArena, a separate family-fit check outside this PMLB panel, is not pooled with the density, neural, or Forest-Flow blocks. An unexecuted method is not a win for DOPE. Later sections point here instead of repeating this list.

## II. RELATED WORK

Tabular synthesis has moved from parametric copulas [2] and Chow–Liu trees [6] through conditional GANs, including TableGAN, and variational autoencoders [1], [7] to diffusion and score-based models [4], [8]–[14]. Autoregressive and language-model generators [15], [16], prior-data fitted generators [17], [18], CTAB-GAN and CTAB-GAN+ [19], [20], and graphical neural generators [21] target the same mixed-type tables. ARF estimates density with a forest [3]. CART synthesis is the synthpop line [5]. A recent survey collects these design choices [22]. This paper runs the Gaussian copula, a Chow–Liu baseline written for this comparison, independent marginals written for this comparison, CTGAN, TVAE, and a 6-lineage Forest-Flow confirmation. New retained panels also measure ARF and a scaled author-default TabSyn subset (Section V-D). TabDDPM on the 100-lineage panel, STaSy, CoDi, TabDiff, CTAB-GAN+, GReaT, REaLTabFormer, and TabPFGen are cited and not measured. SMOTE [23] was not run. This panel scores regression retention, not classification oversampling.

Differentially private Bayesian networks and marginal synthesizers [24]–[27] optimize a privacy budget. Membership inference and shadow-model attacks are a separate detection literature [28]–[30]. Formal differential privacy is a different claim

from an empirical resemblance score [31], [32]. Utility reviews separate marginal fidelity from downstream prediction [33], [34]. Alpha-precision, beta-recall, SDMetrics, and synthcity are the usual fidelity reports [35]–[37]. The fidelity results add alpha-precision, beta-recall, a distance-to-closest-record share, and a DOMIAS AUC. ARF and the retained TabSyn subset have an additional fidelity, detection and empirical-privacy panel in Section V-D. The displayed-profile ARF publication median is 0.840. The pre-specified privacy-attack panel remains not measured. Nothing in those fields is a formal privacy guarantee or a HIPAA de-identification claim.

Tree models remain strong tabular predictors, which is why CatBoost is one of the three auditors [38]–[40]. TabZilla and TabArena show how brittle a single split can be [41], [42]. The regression lineages come from PMLB [43], [44]. Comparing several methods on the same lineages needs a rank test and a critical difference, or a Bayesian signed-rank alternative [45], [46]. This paper reports the frequentist procedure: a Friedman test and a Nemenyi critical difference on the common set, plus paired Wilcoxon tests. The Bayesian signed-rank procedure is not computed.

### III. METHOD

#### A. Displayed generator

The displayed profile is the generator this paper reports. Its configuration id is `features12_steps2048`: at most 12 screened inputs, a hidden width of 16, a tanh activation, 2,048 AdamW steps at learning rate 0.002, a refit readout, and no embedding. It is a compact neural residual, not a GAN, a diffusion model, or a transformer. Fig. 1 is the pipeline. Training screens the inputs, maps them through the width-16 tanh layer, and runs AdamW on the regression mean-squared error, with gradient clipping at 5. The hidden weights and biases are then frozen. The intercept, the linear skip, and the hidden readout are refit on those frozen features. Coefficients whose absolute value is below  $10^{-5}$  are dropped from the skip. For a selected input vector  $x_S$ ,

$$h_j = \tanh(b_j + w_j^\top x_S), \quad \hat{y} = a + \beta^\top x_S + v^\top h. \quad (1)$$

There is no embedding table in (1). A separate dataset-action router uses a vector of dimension  $856 + 24 \times (128 + 10) = 4,168$ . That vector does not generate the rows in this paper.

Equation (1) is the conditional mean, not the joint draw. Sampling for this profile uses the symbolic kernel stored in the artifact. It does not use the separate neural-joint sampler. The dependence stored with an artifact is one choice from the compile search: independence, a Chow–Liu tree, a triangular autoregression, a sparse Gaussian copula, a factor model, a truncated vine, or a two-component mixture. This paper does not treat the displayed artifacts as one dependence family.

Training fits the residual. Sampling draws a row from the stored artifact. Training, not sampling, screens inputs by absolute rank correlation with the target and keeps at most 12. The artifact stores a marginal and a transform for each feature, the selected dependence, the residual, a noise model, and a per-feature missingness rate. A draw samples a latent vector from the stored dependence, maps each coordinate through

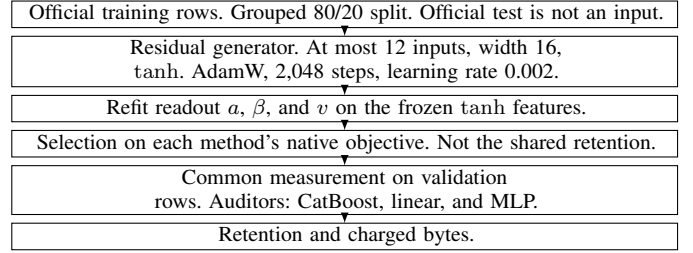


Fig. 1. Displayed pipeline. The official test is not an input.

the standard normal distribution function, inverts the marginal, and inverts the feature transform. The residual then supplies  $\hat{y}$ . When the training target is not entirely zeros and ones, the stored noise is heteroscedastic: a linear fit of the log absolute residual on up to four screened features, with the scale clamped between the lower and upper deciles of those absolute residuals. The sampled target adds Gaussian noise of that scale and then clamps the sum to the unit interval of the training transform. It is not returned in the original units of the source column. Each feature is then replaced, independently, by a missing mark with its stored probability. The charged byte count is the length of the encoded artifact. Before that encoding, the compiler quantizes the marginals, the dependence, the residual, and the noise model.

The profile was fixed before the shared retention was inspected. Compile-time search records the proxy  $0.45u + 0.25d + 0.30j$  and then keeps the candidate with the smallest normalized gate shortfall, then the smallest artifact, then the lowest candidate id. Every candidate record withholds a release certificate. The compile record is not a release certificate. No global DOPE family has been selected. The production default, used when this research profile is not selected, clamps the hidden width from 4 to 12 and trains for 96 steps. It is not this result. Other shapes exist and are not this measurement: micro-TVAEs, TinyMAT, joint TVAE, a masked transformer, and a TabSyn candidate. A direct-rank TabDDPM candidate is a research comparator with no claim on the 10,240-byte tier. The scored runs did not store a per-step loss. A later logged replay did. The supplement plots that replay. The replay files are not substituted for the scored artifacts, and a lower hidden-basis loss at 8,192 steps does not replace the displayed profile. The full width-and-step grid is not measured.

#### B. A fixed encoding of the columns

The displayed generator and the dataset encoding both reduce a table to a fixed set of columns. The budgets differ, and both budgets belong to the current method.

Training screens columns by absolute rank correlation with the target and keeps at most 12. Those columns are the support  $x_S$  in (1). The artifact stores their positions in the training table. That screen is what keeps the residual inside the byte cap. It is a weak dataset key. A column the target does not rank is dropped, and a permutation of the remaining columns changes the stored positions.

The encoding an index can store is the sketch the router already consumes. `embed-dataset` min-max normalizes each training column onto the unit interval, leaves a missing entry missing, and does not read the official test. Each column becomes twelve content summaries: a missing rate, moments, quantiles, a unique-value share, a tail ratio, and the absolute association with the target. Columns are sorted by those summaries, variance first, then by the rest of the summary, then by the sorted observations. A permutation of columns or of rows does not move the vector. The sketch keeps the first 64 columns in that order and pads a shorter table with zeros. Absolute correlations among the kept columns collapse to one six-number distribution, so the width does not grow with the square of the column count. A ten-number header records the task, the log row count, the log column count, and the shape of the target. The width of that sketch is 856.

The 4,168-dimensional vector appends 24 candidate states from one router bundle onto that sketch. A second bundle moves the same table. An index stores the 856-wide sketch. The longer vector is a route from one bundle.

A neighbor found with the sketch is still fit with the displayed profile. The artifact is still charged in bytes, and it still faces the release score in Section III-D. The sketch is not a seventh fitness component. A router action that names a retention or a failure probability is a prediction from the bundle. It does not enter (3).

### C. Retention

Let  $L_{\text{null}}$ ,  $L_{\text{real}}$ , and  $L_{\text{synthetic}}$  be the auditor's loss on held-out validation rows when that auditor is trained on a null baseline, on real fit rows, and on synthetic rows. Those held-out rows come from the official training file. Retention is

$$\frac{L_{\text{null}} - L_{\text{synthetic}}}{L_{\text{null}} - L_{\text{real}}}. \quad (2)$$

Rearrangement, with  $r$  the retention, is algebra on those three losses:

$$L_{\text{synthetic}} = L_{\text{real}} + (1 - r)(L_{\text{null}} - L_{\text{real}}).$$

Invented losses 1, 0.01, and retention 0.94 give a synthetic-trained loss of 0.0694, which is 6.94 times the real-trained mean squared error. The CatBoost median in this paper is not this example. A value of 1 means the synthetic-trained auditor matched the real-trained auditor. A value above 1 means it beat that auditor. A negative value means it did worse than the null. A lineage is informative when the real improvement is at least 0.01 times the absolute null loss and is not zero. Otherwise the lineage is omitted. It is not entered as zero. The three auditors are CatBoost [39], ordinary linear regression, and an MLP. The lineage score is the median of sample seeds 101, 211, and 307. The panel score is the median of those lineage scores. Intervals below are percentile lineage-bootstrap intervals, 10,000 resamples of lineages, with the three seeds kept inside the lineage median. They are not per-seed whiskers.

### D. The release score

The decision this project is built to make is whether an artifact may ship. Retention is the measurement this paper

can finish. It is the shared column after each method has been selected on its own objective. A high retention means an auditor trained on synthetic rows recovered the improvement that the same auditor obtained from real fit rows. That can be true of an artifact that is too large to ship, that misses the joint distribution, that copies training rows, or that moves the features a later model relies on. The release score exists so those failures are not traded against a utility median.

MFS-v2 is that score. With component  $c_i \in [0, 1]$ , weight  $w_i > 0$ ,  $\epsilon = 10^{-6}$ , and  $W = \sum_i w_i$ ,

$$\text{MFS-v2} = 100 \exp\left(\frac{1}{W} \sum_i w_i \ln(\epsilon + c_i)\right) \quad (3)$$

when the hard-gate map is non-empty, every hard gate passed, and every positive-weight component is finite and inside  $[0, 1]$ . Otherwise the scalar is null. The mean is geometric. A component near zero pulls the product down. A component that is missing, or that falls outside  $[0, 1]$ , removes the score rather than publishing a small number. An empty gate map, or any failed hard gate, does the same. Privacy has soft weight zero. A membership failure, an attribute failure, or a copied row is a gate. Utility cannot offset it. The weights are utility transfer 0.30, driver fidelity 0.20, distribution fidelity 0.20, structure fidelity 0.15, coverage realism 0.10, and compactness 0.05.

The six components ask six different questions. Utility transfer is a one-sided lower bound on retention, clamped into  $[0, 1]$ . It is the bound inside (3), and it is distinct from the median of (2) in the figures below. Driver fidelity is an equal mixture of feature-importance rank agreement and top-feature-set overlap between the real table and the synthetic table. Distribution fidelity is half the minimum marginal fidelity, plus a quarter of the sliced-Wasserstein fidelity and a quarter of the maximum mean discrepancy fidelity. Structure fidelity is the minimum dependence fidelity. Coverage realism, on a fixed embedding of holdout rows, is 0.25 precision, 0.25 recall, 0.20 density capped at 1, and 0.30 coverage. Compactness is  $\text{clamp}(1 - \text{bytes}/10240, 0, 1)$ . An artifact past 10,240 bytes fails the byte gate, and the scalar stays null. Crossing the cap is a failed gate, distinct from a retention of zero.

PTF-v1 is the other release KPI. Its hard gates include joint fidelity, membership and attribute attacks, copy checks, and feature-importance agreement on a sealed holdout. It is null here, as is MFS-v2. Compile-time search records the proxy  $0.45u + 0.25d + 0.30j$  and then keeps the candidate with the smallest gate shortfall, then the smallest artifact. That record withholds a release certificate. The proxy is not (3).

No artifact in this paper has a measured MFS-v2 value. Emitting one requires the six components, a non-empty hard-gate map that passed, both the  $\text{size-}n$  and  $\text{size-}4n$  artifacts, real-versus-real controls, artifact-only sampling, and locked metric implementations. A retention win does not supply those inputs. Copying a retention number into (3) would publish a value the contract refuses to emit. The supplement applies (3) to invented components, including a zero compactness term. Those rows are arithmetic. They show the shape of the product. They are not a score for any method in the tables. Baselines

stay on their native objectives. The shared columns in this paper stay retention and charged bytes.

#### IV. EXPERIMENTAL SETUP

##### A. Partitions and provenance

The panel is the rights-cleared PMLB regression collection [43], [44]. Of these prepared lineages, 20 names begin with feynman, which the PMLB catalog describes as synthetic physics, 14 begin with strogatz, which that catalog describes as simulated dynamics, 25 contain the Friedman-style fri marker, 6 contain bng and are Bayesian-network generators, and 35 are the remainder. This paper does not re-verify which of the remainder are observational measurements. Distinct lineage identifiers are not a claim of independent source families. Related names, including the cpu\_small lineages, stay inside the one-row-per-lineage bootstrap. The resampling unit is the lineage, so related variants move together only by chance. A source-family interval is not published, because a verified parent map is not in the ledger. The data lock admits 104 MIT-licensed entries and prepares 100. 4 lineages have no split object because the same source-row digest appears in two partitions (check\_nonoverlap raises on that overlap). Another 3,755 catalog entries lack a redistribution grant and are not backfilled. The prepared cut is a grouped 80/20 of the official training rows, seed 1729. Worker directories hold train.csv and validation.csv. They do not hold test.csv. Official tests stay sealed, and final evaluation is not authorized. The fit view has 72 to 800 rows. Sources above the quota use a hybrid coreset that keeps the observed feature and target range. The supplement records that draw. Held-out validation rows are the grouped holdout of the official training file, and those source rows include the simulated generators named above. Table I is the lock and the file-name count. Test-file contents were not read.

Fit seed 11 is the population actually measured. Sizes  $n$  and  $4n$  are both stored. The tables below are  $4n$ . The Gaussian copula row is Copulas 0.14.1 GaussianMultivariate, selected by held-out mean log density. CTGAN and TVAE are the author implementation, selected by the mean of linear and MLP training-mean skill on validation rows, the same skill defined in Section V. Chow-Liu uses eight bins and Laplace smoothing 1.0. It is an implementation written for this comparison, as are the independent marginals. The ARF watch log's last count is 799 closed and 799 ok out of 800 planned, then a sealing record. The later matched ARF publication supplies scored retention; its displayed-profile comparison median is 0.840. The new retained diagnostic panel has a separate receipt and aggregation rule. Forest-Flow on 6 lineages uses the author regression objective on its frozen grid. Native objectives are not ranked across methods.

##### B. Statistics

Comparisons are paired on the lineages where both methods have a complete informative group. The tests are two-sided Wilcoxon signed-rank tests on the nonzero paired differences, and one-sided sign tests for a positive difference, with a matched-pairs rank-biserial correlation. Holm correction is

applied inside a family, separately for each test, and is not pooled across families. On CatBoost the Holm sign-test values are  $2.15 \times 10^{-15}$ ,  $6.72 \times 10^{-24}$ , and  $4.95 \times 10^{-27}$ . The density family has nine tests (three comparators times three auditors). The neural family is CTGAN and TVAE times the auditors. The Forest-Flow family is separate and has six lineages. A Friedman test and a Nemenyi critical difference use the four density methods on their common complete-informative set, with rank 1 assigned to the higher retention. The critical-difference constant for four methods at  $\alpha = 0.05$  is 2.569. The interval resamples the informative lineages of this panel. Each lineage contributes the median of sample seeds 101, 211, and 307 from the single fit at seed 11. The interval does not include a new fit, a new split, a new population, or a change of auditor. These are descriptive validation tests. They are not the release KPI in Section I-0a, and they do not authorize a release at the 10,240-byte tier. MFS-v2 and PTF-v1 stay null.

##### C. Compute

Table II lists fit time, peak memory, the GPU models, GPU hours, and charged bytes for receipts that store a fit timer. Sample-timer medians (maxima) are 186.7 (197.0) for CTGAN, 186.1 (196.6) for TVAE, and 342.4 (511.7) for Forest-Flow. The DOPE receipt stores no sample timer. No receipt stores a separate CPU timer. Gaussian copula, Chow-Liu, independent marginals, and ARF have no fit timer, so they are omitted; their byte medians are in Table VI. ARF receipts do not record a GPU model. TabSyn and TabDDPM costs are not measured and not measured. The supplement states the replay walls, the Forest-Flow sums, and the unsplit density attempt timers.

#### V. RESULTS

Table III keeps, for each block and auditor, the tuned baseline with the highest paired median retention. The difference is DOPE minus that baseline. Its interval is the paired lineage bootstrap already used in the block tables. W/T/L counts lineages with a positive, zero, or negative difference. Holm  $p$  is the adjustment inside that block's existing family. It is not a new family, and the blocks are not pooled. A positive median with a small  $p$ -value still includes lineage losses. The paired median difference is not the difference of the two medians. 0.224 is not 0.940 minus 0.656.

Fig. 2 shows the level of that shared utility under CatBoost. Retention is the fraction of a real-trained auditor's improvement over a null that an auditor trained on synthetic rows recovers. Higher is better. The value is not MFS-v2. MFS-v2 stays null on every method. Each bar is the median on the paired lineages in that block, with the lineage-bootstrap interval. The two panels are not pooled. Forest-Flow is a separate 6-lineage block. Its CatBoost median is higher than the paired DOPE median on that block, the Holm test is not significant, and that block is not drawn here.

##### A. Density block

Fig. 3 plots retention against  $\log_{10}$  charged bytes for the three auditors. Circles are DOPE, squares the Gaussian copula,

TABLE I  
TEST-SPLIT PROVENANCE. FILE-NAME COUNTS ONLY.

| Item                                               | Record                                                                 |
|----------------------------------------------------|------------------------------------------------------------------------|
| Catalog SHA-256                                    | ab9fda8d2dea46067b70a42812e9d3c1d9dc6ba025780100df34377e81aa1120       |
| Eligible / prepared / overlap exclusions           | 104 / 100 / 4                                                          |
| Unknown-rights catalog entries held out            | 3755                                                                   |
| Split                                              | official_test_grouped_training_80_20, seed 1729                        |
| Overlap lineage ids                                | 590210721973399a, 9fe58a656fdb458a, d06a584f07d0acd5, d0ba629ed07b5d63 |
| Final evaluation authorized                        | false                                                                  |
| S3 worker directories / train.csv / validation.csv | 100 / 100 / 100                                                        |
| test.csv under S3 workers                          | 0                                                                      |
| BeyondArena workers / worker test.csv              | 5 / 0                                                                  |
| Evaluator test.csv files, contents not read        | 5                                                                      |

TABLE II  
FIT TIME, MEMORY, GPU, AND CHARGED BYTES WHERE A FIT TIMER WAS STORED.

| Method      | Fit s        | RAM / VRAM   | Hardware                                                         | GPU h | Bytes                                    |
|-------------|--------------|--------------|------------------------------------------------------------------|-------|------------------------------------------|
| DOPE        | 7.25(13.5)   | 920.7/ 1067  | one NVIDIA GeForce RTX 4090 and two NVIDIA GeForce RTX 3090 GPUs | 0.193 | 1,895.5(758–22,233)                      |
| CTGAN       | 213.5(268.1) | 1557.8/ 1101 | NVIDIA GeForce RTX 4090                                          | 2.52  | 741,834(528,075–1,428,695)               |
| TVAE        | 214.2(253.2) | 1252.9/ 1069 | NVIDIA GeForce RTX 4090                                          | 2.45  | 317,141.5(157,578–741,659)               |
| Forest-Flow | 185.9(380.6) | 6112.0/ 3181 | NVIDIA GeForce RTX 4090                                          | 2.44  | 325,848,913.5(115,202,253–1,085,155,062) |

TABLE III  
STRONGEST TUNED BASELINE AT  $4n$ . W/T/L COUNTS LINEAGES. HOLM STAYS INSIDE THE BLOCK.

| Block   | Auditor  | Baseline        | $n$ | Difference                | W/T/L   | Holm $p$               |
|---------|----------|-----------------|-----|---------------------------|---------|------------------------|
| Density | CatBoost | Gaussian copula | 97  | 0.224 [0.149, 0.290]      | 86/0/11 | $1.35 \times 10^{-13}$ |
| Density | Linear   | Gaussian copula | 92  | 0.00366 [-0.00438, 0.012] | 49/0/43 | 0.373                  |
| Density | MLP      | Gaussian copula | 75  | 0.185 [0.128, 0.228]      | 65/0/10 | $4.96 \times 10^{-11}$ |
| Neural  | CatBoost | TVAE            | 21  | 0.236 [0.087, 0.431]      | 19/0/2  | $1.57 \times 10^{-4}$  |
| Neural  | Linear   | TVAE            | 21  | 0.263 [0.166, 0.410]      | 20/0/1  | $1.57 \times 10^{-4}$  |
| Neural  | MLP      | TVAE            | 19  | 0.284 [0.153, 0.734]      | 17/0/2  | $1.57 \times 10^{-4}$  |
| Forest  | CatBoost | Forest-Flow     | 6   | -0.027 [-0.308, 0.00804]  | 1/0/5   | 0.375                  |
| Forest  | Linear   | Forest-Flow     | 5   | 0.012 [-0.260, 0.302]     | 3/0/2   | 0.812                  |
| Forest  | MLP      | Forest-Flow     | 4   | -0.032 [-0.227, -0.010]   | 0/0/4   | 0.375                  |

triangles Chow–Liu, and diamonds the independent marginals. Filled marks are lineages. Open marks are that method’s median and its lineage-bootstrap interval on the lineages it completed. The table intervals are the paired subsets. For CatBoost those two DOPE sets coincide. Paired differences outside  $[-1.5, 1.5]$  are counted in the panel title and omitted from that frame. The retention axis stops at 1.6. The dashed line is the cap. Neural and Forest-Flow artifacts are not on this figure.

Table IV is the paired density block. Under CatBoost, DOPE’s median paired differences are positive against the Gaussian copula (0.224, interval 0.149 to 0.290), the Chow–Liu tree, and the independent marginals. The Holm-adjusted  $p$ -values are  $1.35 \times 10^{-13}$ ,  $4.85 \times 10^{-16}$ , and  $1.17 \times 10^{-16}$ . The rank-biserial correlations are 0.890, 0.978, and 0.999. The Gaussian copula is the strongest density baseline. Wins, ties, and losses against it are 86/0/11 under CatBoost and 65/0/10 under the MLP. On the 97 lineages common to all four methods, the Friedman  $p$ -value is  $2.71 \times 10^{-44}$  and the critical difference is 0.476. All six CatBoost pairs exceed that difference. Fig. 4 shows the sorted paired differences with the same four marks. The horizontal axis is the retention difference. The segment is the critical difference.

The linear auditor does not separate DOPE from the Gaussian copula (49/0/43; Holm 0.373; difference 0.00366, interval

–0.00438 to 0.012; rank-biserial 0.107). The interval contains zero. That non-separation is not an equivalence test. Ranks are 1.533 and 1.598 against critical difference 0.489. Chow–Liu and the independent marginals remain separated under the linear auditor and the MLP. A CatBoost separation is not a separation under every auditor.

Table V is the median raw loss on the same informative lineages. RMSE is the square root of the stored mean squared error. Training-mean skill is one minus that loss divided by the training-mean null loss. Conventional held-out  $R^2$  uses the mean of the evaluation targets and was not stored for these rows. Retention is the median of the per-lineage ratios, not the ratio of these medians. CatBoost reads RMSE 0.053 and 0.081, with training-mean skill 0.881 and 0.792.

Table VI counts artifacts at or under the cap. The two displayed DOPE artifacts over the cap are 4544\_Geographical OriginalofMusic at 22,233 bytes and 588\_fri\_c4\_1000\_100 at 19,019 bytes. 678\_visualizing\_environmental has a fit whose CatBoost group is not informative, so the CatBoost median uses 97 lineages while the byte count uses 100. Independent marginals pass the cap more often than the Chow–Liu baseline and still sit near retention zero. Smallness alone is not the result. The displayed byte range runs from 758 to 22,233.

On the density cells, the displayed profile has median classifier two-sample AUC 0.517 (interval 0.504 to 0.532,

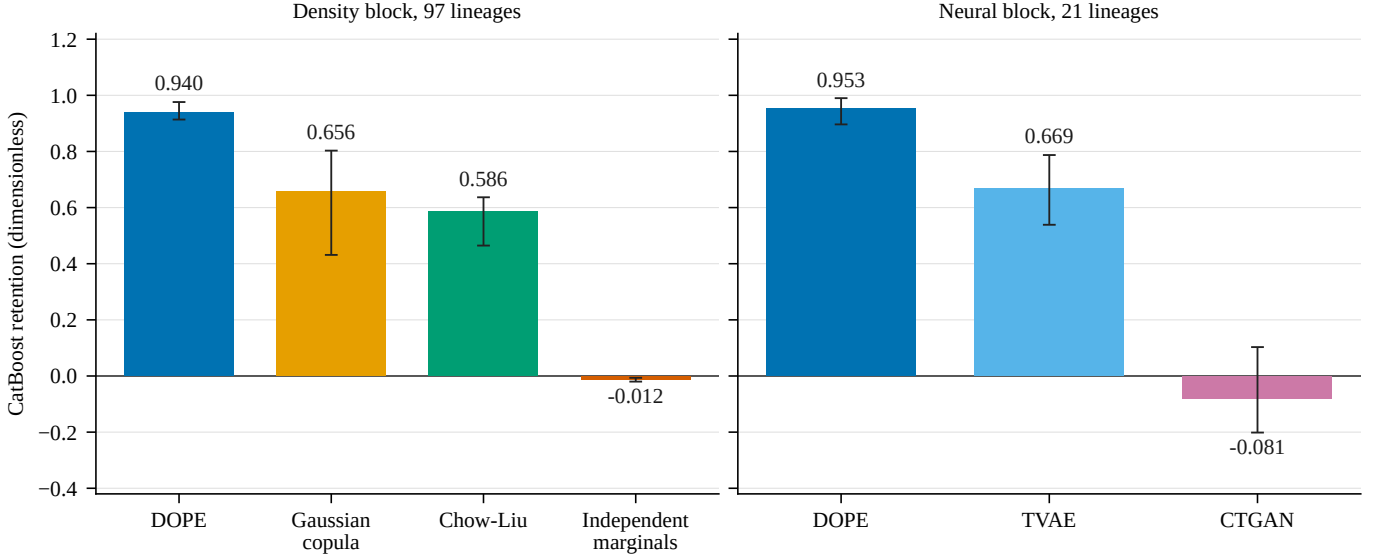


Fig. 2. CatBoost retention at  $4n$ . Higher is better. Each bar is the paired-lineage median and its lineage-bootstrap interval. Density uses 97 lineages. The neural panel uses 21 and is not pooled. Forest-Flow, 6 lineages, is higher on its own block and is omitted.

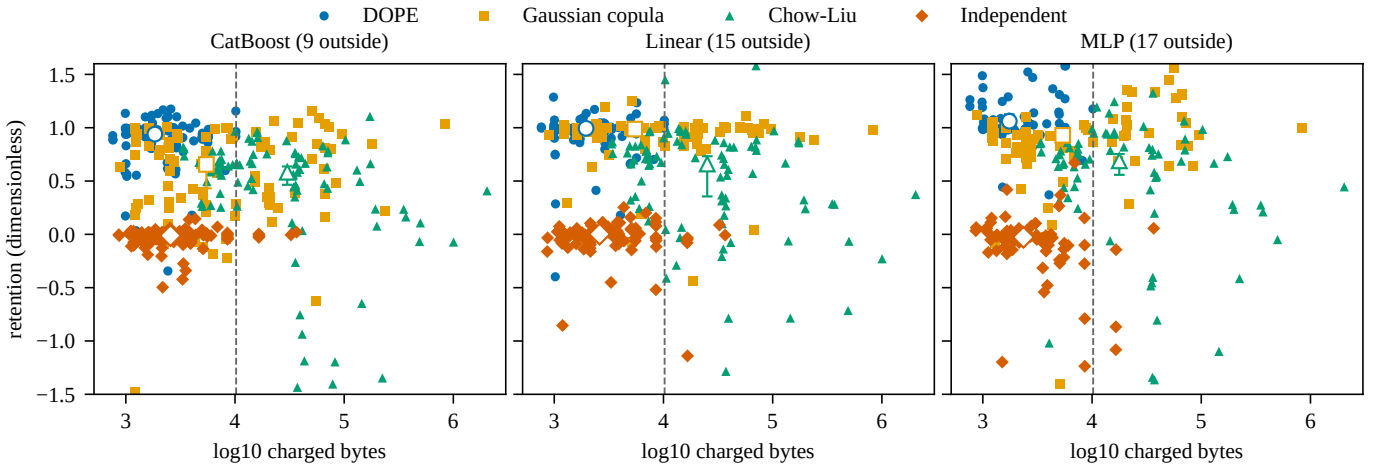


Fig. 3. Density-block retention at  $4n$  against charged bytes. Three auditors. The dashed line is the 10,240-byte cap.

$n = 98$ ), mean marginal Kolmogorov–Smirnov distance 0.114 (0.103 to 0.135), and pairwise correlation fidelity 0.953 (0.945 to 0.961). A two-sample AUC near one half means a linear-time classifier has trouble telling the synthetic rows from the held-out validation rows. Table VII adds alpha-precision, beta-recall, the unbalanced DCR share, and a DOMIAS AUC. The lower block is the neural lineages only. Holm families are 12 and 8, separate from retention. The DCR column is the fraction of synthetic queries whose nearest reference is in the fit view, with ties scored one half. A balanced variant was computed in the harness and is not this column. Under an independent draw, the unbalanced share sits near the fit-view fraction of the pooled references. The grouped split keeps about four fifths of those rows in the fit view, so the baseline is near four fifths, not one half. Higher values of that share, and higher DOMIAS AUC, sit nearer the fit view and are not a quality win. DOMIAS here is the isolated kernel density-ratio equation

from the pinned author expression, evaluated on this numeric projection. It is not the original DOMIAS training pipeline and it is not a membership-inference certificate. The DOMIAS AUC stays near one half. The density DCR contrast against independent marginals clears Holm in that worse direction. The displayed-profile ARF publication median is 0.840. The pre-specified privacy-attack panel remains not measured. This is not differential privacy or HIPAA de-identification.

### B. Neural block

Table VIII is not pooled with Table IV. On the 21 CatBoost lineages, DOPE’s median is 0.953, CTGAN’s is  $-0.081$ , and TVAE’s is 0.669. TVAE is the stronger neural baseline. Wins, ties, and losses against TVAE are 19/0/2, 20/0/1, and 17/0/2. The median favors DOPE, and DOPE still loses lineages. Every successful CTGAN and TVAE artifact in this byte count is over the cap (0/22 and 0/22). Runs that stopped at the recorded

TABLE IV  
DENSITY BLOCK AT  $4n$ . W/T/L COUNTS LINEAGES.

| Auditor  | Comparator       | $n$ | DOPE                 | Other                     | Difference                | W/T/L   | Holm $p$               |
|----------|------------------|-----|----------------------|---------------------------|---------------------------|---------|------------------------|
| CatBoost | Gaussian copula  | 97  | 0.940 [0.914, 0.976] | 0.656 [0.431, 0.803]      | 0.224 [0.149, 0.290]      | 86/0/11 | $1.35 \times 10^{-13}$ |
| CatBoost | Chow-Liu         | 97  | 0.940 [0.914, 0.976] | 0.586 [0.465, 0.637]      | 0.337 [0.282, 0.451]      | 94/0/3  | $4.85 \times 10^{-16}$ |
| CatBoost | Indep. marginals | 97  | 0.940 [0.914, 0.976] | -0.012 [-0.020, -0.00649] | 0.956 [0.918, 0.980]      | 96/0/1  | $1.17 \times 10^{-16}$ |
| Linear   | Gaussian copula  | 92  | 0.994 [0.981, 0.996] | 0.989 [0.968, 0.994]      | 0.00366 [-0.00438, 0.012] | 49/0/43 | 0.373                  |
| Linear   | Chow-Liu         | 92  | 0.994 [0.981, 0.996] | 0.661 [0.356, 0.738]      | 0.340 [0.244, 0.425]      | 86/0/6  | $7.12 \times 10^{-15}$ |
| Linear   | Indep. marginals | 92  | 0.994 [0.981, 0.996] | 0.00206 [-0.023, 0.00583] | 0.988 [0.975, 1.001]      | 92/0/0  | $5.70 \times 10^{-16}$ |
| MLP      | Gaussian copula  | 75  | 1.061 [1.024, 1.125] | 0.928 [0.865, 0.953]      | 0.185 [0.128, 0.228]      | 65/0/10 | $4.96 \times 10^{-11}$ |
| MLP      | Chow-Liu         | 75  | 1.061 [1.024, 1.125] | 0.692 [0.582, 0.754]      | 0.530 [0.350, 0.651]      | 74/0/1  | $2.11 \times 10^{-13}$ |
| MLP      | Indep. marginals | 75  | 1.061 [1.024, 1.125] | -0.028 [-0.047, -0.013]   | 1.104 [1.047, 1.179]      | 75/0/0  | $2.11 \times 10^{-13}$ |

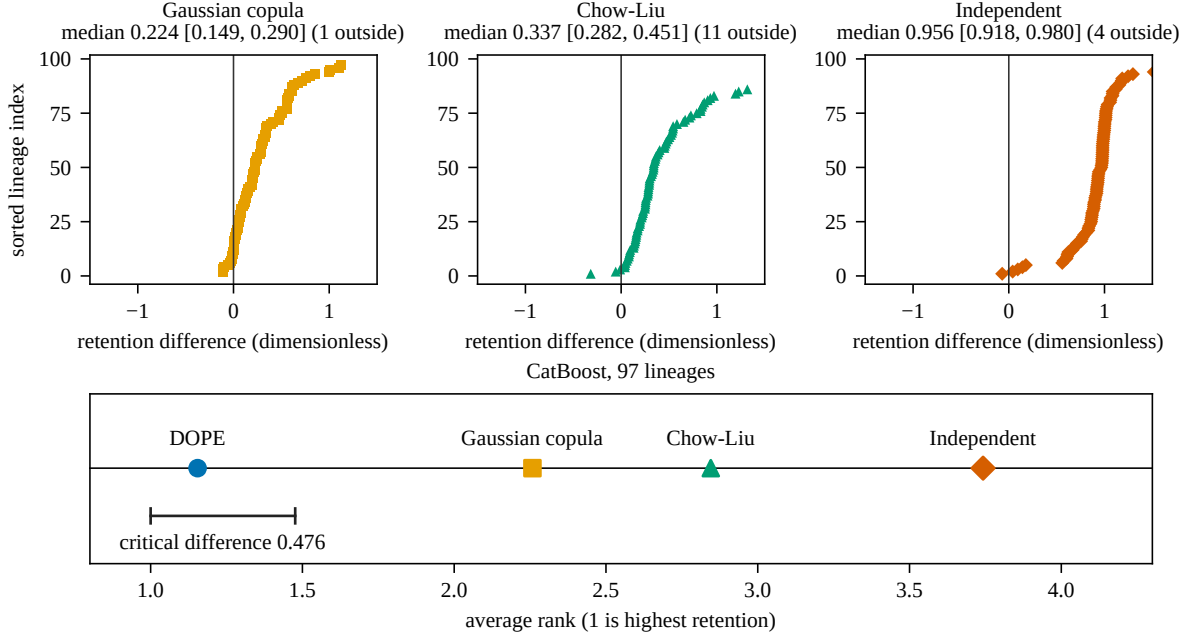


Fig. 4. CatBoost paired differences, sorted, and the Nemenyi scale. Rank 1 is the higher retention. The segment length is the critical difference.

TABLE V  
RAW AUDITOR MEDIAN ON THE INFORMATIVE LINEAGES.

| Auditor  | $n$ | RMSE real | RMSE synth. | training-mean skill, real | training-mean skill, synth. |
|----------|-----|-----------|-------------|---------------------------|-----------------------------|
| CatBoost | 97  | 0.053     | 0.081       | 0.881                     | 0.792                       |
| Linear   | 92  | 0.105     | 0.112       | 0.585                     | 0.551                       |
| MLP      | 75  | 0.098     | 0.083       | 0.666                     | 0.785                       |

TABLE VI  
CHARGED BYTES AT  $4n$ . FOREST-FLOW IS NOT POOLED.

| Method                | Median bytes  | Within cap |
|-----------------------|---------------|------------|
| DOPE reference        | 1,895.5       | 98/100     |
| Gaussian copula       | 5,464.5       | 62/100     |
| Chow-Liu              | 29,865        | 31/100     |
| Independent marginals | 2,549.5       | 94/100     |
| CTGAN                 | 741,834       | 0/22       |
| TVAE                  | 317,141.5     | 0/22       |
| Forest-Flow           | 325,848,913.5 | 0/6        |
| ARF                   | 1,627,718.5   | 0/100      |

time limit are omitted. They are not counted as failures of a finished fit.

### C. Forest-Flow

Table IX is a 6-lineage confirmation, not pooled with the density or neural blocks. The Forest-Flow CatBoost median is 1.003 (interval 0.926 to 1.026) and the paired DOPE median is 0.940. Wins, ties, and losses are 1/0/5 under CatBoost, 3/0/2 under the linear auditor, and 0/0/4 under the MLP. DOPE is lower on every paired MLP lineage. The Holm tests are not significant at this count, so this is not a certified win for either method. Charged Forest-Flow artifacts run from 115,202,253 to 1,085,155,062 bytes (median 325,848,913.5) against a DOPE median of 2,067 on the same lineages. MFS-v2 on this block is null.

Table X is the published matched-population receipt for ARF at  $4n$  on the displayed profile. Complete-population TabDDPM, TabSyn, and Forest-Flow receipts are omitted; the completed retained TabSyn subset is reported separately below.

### D. Retained validation evidence

Tables XI and XII add the merged retained-sample evidence. Each lineage averages all three sample seeds before a lineage-bootstrap median and percentile interval are computed. This

TABLE VII  
GROUPED-VALIDATION FIDELITY AND EMPIRICAL PRIVACY. NOT DP OR HIPAA.

| Method   | $\alpha$ -prec.<br>Median [interval] (lineages). | $\beta$ -recall            | DCR share<br>(unbalanced)  | DOMIAS                    |
|----------|--------------------------------------------------|----------------------------|----------------------------|---------------------------|
| DOPE     | 0.905 [0.889, 0.918] (98)                        | 0.437 [0.398, 0.473] (98)  | 0.811 [0.806, 0.814] (98)  | 0.508 [0.498, 0.520] (94) |
| Gaussian | 0.812 [0.786, 0.832] (100)                       | 0.430 [0.378, 0.453] (100) | 0.806 [0.802, 0.811] (100) | 0.513 [0.494, 0.527] (94) |
| Chow–Liu | 0.858 [0.825, 0.904] (100)                       | 0.359 [0.312, 0.400] (100) | 0.810 [0.806, 0.816] (100) | 0.512 [0.495, 0.522] (94) |
| Indep.   | 0.885 [0.868, 0.897] (100)                       | 0.284 [0.197, 0.347] (100) | 0.801 [0.797, 0.806] (100) | 0.507 [0.496, 0.521] (94) |
| DOPE     | 0.902 [0.882, 0.940] (22)                        | 0.441 [0.381, 0.489] (22)  | 0.811 [0.802, 0.819] (22)  | 0.508 [0.480, 0.538] (22) |
| CTGAN    | 0.719 [0.520, 0.856] (22)                        | 0.170 [0.142, 0.267] (22)  | 0.820 [0.800, 0.842] (22)  | 0.502 [0.463, 0.534] (22) |
| TVAE     | 0.850 [0.731, 0.896] (22)                        | 0.356 [0.283, 0.401] (22)  | 0.825 [0.813, 0.843] (22)  | 0.527 [0.487, 0.563] (22) |

| Contrast                                                            | $\alpha$ -prec.                                                    | $\beta$ -recall                                                                     | DCR share<br>(unbalanced)                                                  | DOMIAS                                                                    |
|---------------------------------------------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Line 2: Holm $p$ ; rank-biserial $r$ ; $n$ . DOPE minus comparator. |                                                                    |                                                                                     |                                                                            |                                                                           |
|                                                                     | 0.083 [0.045, 0.113]                                               | 0.024 [0.011, 0.039]                                                                | -0.0025 [-0.00625, 0.005]                                                  | 0.00157 [-0.0069, 0.0074]                                                 |
| DOPE–Gaussian                                                       | $1.86 \times 10^{-10}$ ; 0.781; 98<br>0.019 [0.00782, 0.056]       | 0.394; 0.215; 98<br>0.048 [0.029, 0.072]                                            | 1.000; 0.000206; 98<br>-0.00125 [-0.00562, 0.00125]                        | 1.000; -0.085; 94<br>0.000581 [-0.00802, 0.011]                           |
| DOPE–Chow–Liu                                                       | $2.00 \times 10^{-5}$ ; 0.551; 98<br>0.018 [0.0072, 0.028]         | $1.32 \times 10^{-10}$ ; 0.789; 98<br>0.104 [0.068, 0.138]                          | 0.806; -0.169; 98<br>0.00625 [0.0025, 0.011]                               | 1.000; 0.00572; 94<br>0.00244 [-0.00595, 0.015]                           |
| DOPE–Indep.                                                         | $5.47 \times 10^{-5}$ ; 0.523; 98                                  | $9.71 \times 10^{-15}$ ; 0.937; 98                                                  | 0.005; 0.398; 98                                                           | 1.000; 0.080; 94                                                          |
| DOPE–CTGAN                                                          | 0.168 [0.054, 0.368]<br>0.004; 0.794; 22<br>0.071 [0.00713, 0.141] | 0.219 [0.131, 0.256]<br>$3.81 \times 10^{-6}$ ; 1.000; 22<br>0.062 [0.00546, 0.135] | -0.00812 [-0.029, 0.015]<br>0.532; -0.277; 22<br>-0.025 [-0.030, -0.00641] | -0.00028 [-0.014, 0.031]<br>0.610; 0.130; 22<br>-0.021 [-0.036, -0.00972] |
| DOPE–TVAE                                                           | 0.040; 0.644; 22                                                   | 0.069; 0.573; 22                                                                    | 0.059; -0.613; 22                                                          | 0.108; -0.510; 22                                                         |

TABLE VIII  
NEURAL BLOCK AT  $4n$ , NOT POOLED. W/T/L COUNTS LINEAGES.

| Auditor  | Comparator | $n$ | DOPE                 | Other                  | Difference           | W/T/L  | Holm $p$              |
|----------|------------|-----|----------------------|------------------------|----------------------|--------|-----------------------|
| CatBoost | CTGAN      | 21  | 0.953 [0.896, 0.990] | -0.081 [-0.202, 0.103] | 0.881 [0.702, 1.150] | 21/0/0 | $5.72 \times 10^{-6}$ |
| CatBoost | TVAE       | 21  | 0.953 [0.896, 0.990] | 0.669 [0.539, 0.787]   | 0.236 [0.087, 0.431] | 19/0/2 | $1.57 \times 10^{-4}$ |
| Linear   | CTGAN      | 21  | 0.999 [0.996, 1.015] | -0.033 [-0.225, 0.080] | 1.049 [0.916, 1.236] | 21/0/0 | $5.72 \times 10^{-6}$ |
| Linear   | TVAE       | 21  | 0.999 [0.996, 1.015] | 0.796 [0.605, 0.867]   | 0.263 [0.166, 0.410] | 20/0/1 | $1.57 \times 10^{-4}$ |
| MLP      | CTGAN      | 19  | 1.026 [0.998, 1.263] | -0.149 [-0.471, 0.083] | 1.079 [0.877, 2.030] | 19/0/0 | $1.53 \times 10^{-5}$ |
| MLP      | TVAE       | 19  | 1.026 [0.998, 1.263] | 0.753 [0.478, 0.852]   | 0.284 [0.153, 0.734] | 17/0/2 | $1.57 \times 10^{-4}$ |

TABLE IX  
FOREST-FLOW CONFIRMATION AT  $4n$ , CATBOOST. NOT POOLED.

| Lineage           | DOPE bytes | DOPE  | Forest-Flow bytes | Forest-Flow |
|-------------------|------------|-------|-------------------|-------------|
| 1193_BNG_lowbwt   | 2,436      | 0.970 | 416,921,273       | 1.004       |
| 503_wind          | 3,846      | 0.996 | 637,056,689       | 1.016       |
| 582_fri_c1_500_25 | 5,670      | 0.909 | 1,085,155,062     | 0.883       |
| 596_fri_c2_250_5  | 1,698      | 0.807 | 234,776,554       | 0.968       |
| feynman_l_12_1    | 1,032      | 0.991 | 123,784,681       | 1.001       |
| strogatz_vdp1     | 984        | 0.581 | 115,202,253       | 1.036       |

TABLE X  
PUBLISHED ARF MATCHED RECEIPTS AT  $4n$ , CATBOOST.

| Publication          | Lineages | DOPE  | Receipt | Difference |
|----------------------|----------|-------|---------|------------|
| ARF, author_default  | 97       | 0.940 | 0.840   | 0.077      |
| ARF, native_selected | 97       | 0.940 | 0.826   | 0.095      |

differs from the median-of-seed rule in the earlier blocks; those blocks and their numbers are preserved. A lineage enters a metric only when all three values exist. An unavailable outcome stays absent. Table XI covers the same prepared population for ARF, copula and Chow–Liu defaults and native-selected configurations. Table XII includes only the complete scaled author-default TabSyn lineages and the classical methods on those identical inputs. It cannot establish a full-population TabSyn result or a native-tuned TabSyn comparison. These

evaluations did not select a configuration.

Table XIII gives exploratory TabSyn-minus-comparator retention differences on shared lineages. The Holm family comprises four metrics (CatBoost retention, mean marginal KS/TV, CatBoost C2ST AUC and distance-MIA AUC) times six classical configurations. Conservative binomial/order-statistic intervals for the median difference use Bonferroni correction over the entire family, conditional on independent lineages. This small cohort requires unbounded simultaneous intervals. Finite tail-adjusted bootstrap ranges are retained only as exploratory descriptions; they do not guarantee familywise coverage. Marginal intervals in Fig. 5 are not simultaneous intervals. The machine-readable reduction includes every diagnostic contrast and all applicability exclusions. A raw AUC difference is not a directional utility or privacy win. Selection of the completed subset, one fit seed and its small lineage count limit this

TABLE XI

RETAINED CLASSICAL VALIDATION AT  $4n$ . MEDIAN [95% LINEAGE-BOOTSTRAP INTERVAL] (INFORMATIVE LINEAGES). NATIVE SELECTION USES EACH METHOD'S OWN OBJECTIVE.

| Method   | Config. | CB retention              | KS/TV                      | C2ST CB AUC                | Distance MIA AUC           |
|----------|---------|---------------------------|----------------------------|----------------------------|----------------------------|
| ARF      | Default | 0.820 [0.787, 0.872] (99) | 0.103 [0.097, 0.114] (100) | 0.604 [0.582, 0.626] (100) | 0.553 [0.538, 0.564] (100) |
| ARF      | Native  | 0.806 [0.736, 0.855] (99) | 0.105 [0.098, 0.114] (100) | 0.610 [0.591, 0.627] (100) | 0.542 [0.536, 0.559] (100) |
| Chow-Liu | Default | 0.643 [0.552, 0.672] (99) | 0.128 [0.106, 0.154] (100) | 0.721 [0.652, 0.773] (100) | 0.513 [0.503, 0.525] (100) |
| Chow-Liu | Native  | 0.566 [0.460, 0.631] (99) | 0.126 [0.105, 0.157] (100) | 0.702 [0.641, 0.761] (99)  | 0.513 [0.504, 0.523] (100) |
| Copula   | Default | 0.594 [0.427, 0.762] (99) | 0.126 [0.094, 0.138] (100) | 0.615 [0.591, 0.671] (100) | 0.514 [0.503, 0.524] (100) |
| Copula   | Native  | 0.641 [0.431, 0.803] (99) | 0.134 [0.125, 0.158] (100) | 0.730 [0.696, 0.779] (99)  | 0.507 [0.494, 0.521] (100) |

TABLE XII

MATCHED RETAINED TABSYN SUBSET AT  $4n$ . SAME INPUTS AND REAL-DATA CONTROLS FOR EVERY METHOD; MEDIAN [95% INTERVAL] (LINEAGES). SEPARATE FROM THE FULL CLASSICAL POPULATION.

| Method   | Config.        | CB retention             | KS/TV                    | C2ST CB AUC              | Distance MIA AUC         |
|----------|----------------|--------------------------|--------------------------|--------------------------|--------------------------|
| ARF      | Default        | 0.839 [0.668, 0.942] (8) | 0.129 [0.084, 0.165] (8) | 0.623 [0.547, 0.969] (8) | 0.555 [0.506, 0.613] (8) |
| ARF      | Native         | 0.752 [0.534, 0.946] (8) | 0.132 [0.084, 0.163] (8) | 0.606 [0.583, 0.812] (8) | 0.527 [0.482, 0.558] (8) |
| Chow-Liu | Default        | 0.746 [0.666, 0.859] (8) | 0.186 [0.068, 0.255] (8) | 0.756 [0.525, 0.999] (8) | 0.522 [0.465, 0.542] (8) |
| Chow-Liu | Native         | 0.655 [0.561, 0.778] (8) | 0.188 [0.087, 0.248] (8) | 0.707 [0.525, 0.999] (8) | 0.499 [0.470, 0.533] (8) |
| Copula   | Default        | 0.643 [0.270, 0.935] (8) | 0.146 [0.067, 0.242] (8) | 0.687 [0.554, 0.987] (8) | 0.510 [0.466, 0.541] (8) |
| Copula   | Native         | 0.680 [0.260, 0.858] (8) | 0.157 [0.097, 0.242] (8) | 0.698 [0.642, 0.987] (8) | 0.493 [0.463, 0.525] (8) |
| TabSyn   | Scaled default | 0.911 [0.685, 0.971] (8) | 0.136 [0.120, 0.157] (8) | 0.670 [0.597, 0.869] (8) | 0.571 [0.547, 0.619] (8) |

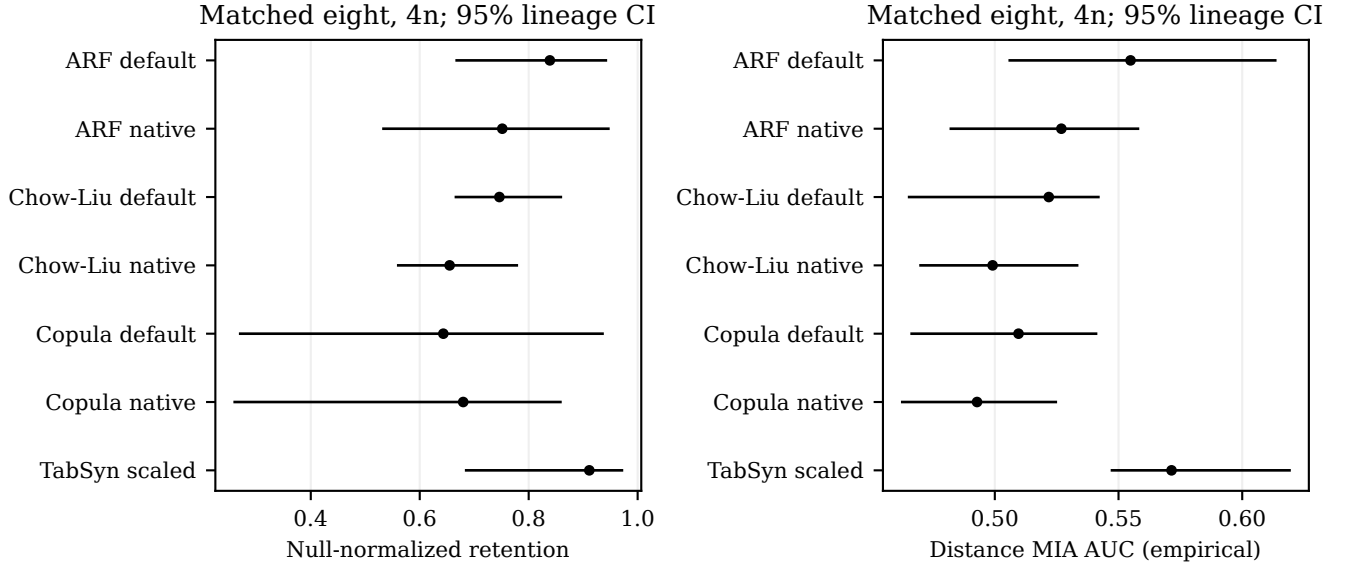


Fig. 5. Matched retained subset: median and marginal lineage-bootstrap intervals. Privacy AUC is an empirical diagnostic; its interval is not a privacy guarantee.

exploratory analysis.

Table XIV separately extends the published DOPE-ARF retention comparison with lineage intervals. It preserves that receipt's median-of-three rule. Its six-test family contains two ARF configurations and three auditors. The shared-lineage intersections differ by auditor. These validation descriptions leave the release and superiority gates null.

The supplement prints the deterministic first-batch diagnostics per physical job. Incomplete seed groups and mixed research profiles prevent population confidence intervals or method rankings for that batch. All reductions trace to committed receipt paths, source digests and JSON pointers in docs/whitepaper/generated/retained-evidence.json. The original receipt references are retained in research/benchmark/results/retained-clas

sical-validation/panel.json and the first-batch publication recorded in that reduction.

The coverage table distinguishes Forest-Flow confirmation from the Forest-Diffusion pilot modes. No full-population diffusion result is inferred from the Flow subset. Additional native-scored fits and published common-metric cells are counted separately. Pending publication does not mean that a method failed. Receipt paths and digests are recorded in docs/whitepaper/generated/baseline-coverage.json. The ARF follow-on now supplies complete five-fit common measurements on the same validation lineages. Its between-fit standard deviation is descriptive, not a confidence interval. TabSyn's additional retained n-sized evaluations include incomplete sample groups; those groups have no aggregate figure value. The supplementary descriptive figure

TABLE XIII

EXPLORATORY MATCHED-SUBSET CATBOOST DIFFERENCES. CONSERVATIVE SIMULTANEOUS MEDIAN INTERVALS AND HOLM  $p$  OVER THE ENTIRE FOUR-METRIC FAMILY; THE CORRECTED INTERVALS ARE UNBOUNDED.

| TabSyn minus     | $n$ | Median CB difference | Simultaneous median CI | Holm $p$ |
|------------------|-----|----------------------|------------------------|----------|
| ARF/default      | 8   | 0.010                | Unbounded              | 1.000    |
| ARF/native       | 8   | 0.159                | Unbounded              | 0.469    |
| Chow-Liu/default | 8   | 0.157                | Unbounded              | 1.000    |
| Chow-Liu/native  | 8   | 0.149                | Unbounded              | 0.188    |
| Copula/default   | 8   | 0.242                | Unbounded              | 1.000    |
| Copula/native    | 8   | 0.245                | Unbounded              | 1.000    |

TABLE XIV

DOPE-ARF LEGACY MATCHED RECEIPT AT  $4n$ : EXPLORATORY PAIRED DIFFERENCES, CONSERVATIVE SIMULTANEOUS MEDIAN INTERVALS AND HOLM  $p$ . MEDIAN-OF-SEED AGGREGATION.

| ARF config. | Auditor  | $n$ | DOPE minus ARF | Simultaneous median CI | Holm $p$              |
|-------------|----------|-----|----------------|------------------------|-----------------------|
| Default     | catboost | 97  | 0.077          | [0.019, 0.154]         | $1.31 \times 10^{-5}$ |
| Native      | catboost | 97  | 0.095          | [0.028, 0.175]         | $5.47 \times 10^{-7}$ |
| Default     | linear   | 92  | 0.027          | [0.0056, 0.090]        | $8.53 \times 10^{-5}$ |
| Native      | linear   | 92  | 0.037          | [0.00582, 0.113]       | $1.62 \times 10^{-5}$ |
| Default     | mlp      | 75  | 0.115          | [0.043, 0.236]         | $4.56 \times 10^{-8}$ |
| Native      | mlp      | 75  | 0.143          | [0.040, 0.253]         | $8.01 \times 10^{-8}$ |

TABLE XV

BASELINE COVERAGE FROM COMMITTED PUBLICATIONS. PENDING ROWS HAVE NO ESTIMATED RESULT OR COMPLETION DATE.

| Method           | Published measured coverage                                                                                                                                                                                  | Remaining campaign |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| TabSyn           | 49 scaled-default lineages, 143 measured cells at fit11: eight complete $n/4n$ lineages (48 cells), plus 95 $n$ cells on 41 lineages (19 complete/22 partial groups). Native tuning/full population pending. | Pending            |
| TabDDPM          | 12 retained lineages; 126 physical/132 logical cells; 21 models, fit seed 11. Default/native cohorts differ; native search incomplete.                                                                       | Pending            |
| Forest-Flow      | 13 default lineages at five fit seeds; 65 models/390 cells. Separate legacy six-lineage default/native panel; full population pending.                                                                       | Pending            |
| Forest-Diffusion | Pilot only; separate 100-lineage matched diffusion panel pending.                                                                                                                                            | Pending            |
| ARF              | 100 default/native lineages at five fit seeds, $n/4n$ , three sample seeds: 6,000 logical cells / 5,892 physical receipts (108 aliases). Between-fit SD available; official-test certification pending.      | Pending            |
| GaussianCopula   | 100 default/native lineages at fit seed 11; 1200 logical cells. Five-fit common coverage pending.                                                                                                            | Pending            |
| Chow-Liu         | 100 default/native lineages at fit seed 11; 1200 logical cells. Five-fit common coverage pending.                                                                                                            | Pending            |

inputs in docs/whitepaper/generated/published-baseline-kpis.csv retain a receipt path and JSON pointer for every value. Cohorts and fit counts differ; these inputs do not support a paired ranking. The ARF validation completion does not admit final official-test certification.

### E. Release score

The scalar in (3) is null for every method in this paper. The supplement shows the invented arithmetic. The evaluation locks for a final test were not frozen.

### F. BeyondArena

BeyondArena (2ecfe882ccfb814fc27c4de10a64ceefd5d7655c) is not one of the 100 PMLB lineages and is not pooled with them.

5 of 12 families fit inside the cap. 7 did not prepare (4 outside the projected range, 3 overlapping). The range was not raised. Table XVI has no retention column. The supplement names all 142 families.

TABLE XVI

BEYONDARENA FAMILIES THAT PREPARED. NO RETENTION AUDITOR.

| Family         | Fit rows | 2,048 B | 8,192 B | Val 2,048 | Val 8,192 |
|----------------|----------|---------|---------|-----------|-----------|
| airfoil        | 802      | 804     | 818     | 0.010     | 0.00448   |
| garments       | 939      | 2,317   | 2,372   | 0.028     | 0.025     |
| musk           | 3,738    | 9,273   | 9,350   | 0.146     | 0.090     |
| telemonitoring | 257      | 1,852   | 1,890   | 0.033     | 0.019     |
| video          | 35,918   | 5,054   | 5,071   | 0.00247   | 0.00195   |

## VI. ABLATIONS

features12\_steps8192 is the same residual with four times as many AdamW steps. It is a budget check, not the displayed model. Where both budgets have a complete three-seed group, the CatBoost median paired difference is 0.034 (68 lineages). The supplement gives the linear and MLP differences and lists all 100 lineages. A blank cell is an unavailable fit or an auditor that is not informative. It is not a zero. The scored fits did not keep a per-step series. The supplement plots the logged replay. It is not a substitute for the scored artifacts. The lower loss at 8,192 steps does not replace the displayed profile.

The primary informativeness threshold is 0.01 times the

TABLE XVII  
INFORMATIVENESS THRESHOLD AND AUDITOR COUNTS.

| Threshold | DOPE       | Gaussian   | Chow-Liu   | Indep.      |
|-----------|------------|------------|------------|-------------|
| 0         | 0.940 (97) | 0.656 (99) | 0.582 (99) | -0.012 (99) |
| 0.001     | 0.940 (97) | 0.656 (99) | 0.582 (99) | -0.012 (99) |
| 0.01      | 0.940 (97) | 0.656 (99) | 0.582 (99) | -0.012 (99) |
| 0.05      | 0.940 (97) | 0.656 (99) | 0.582 (99) | -0.012 (99) |
| 0.1       | 0.942 (96) | 0.668 (98) | 0.584 (98) | -0.012 (98) |
| Threshold | CatBoost   | Linear     | MLP        |             |
| 0.001     | 97         | 93         | 75         |             |
| 0.01      | 97         | 92         | 75         |             |
| 0.05      | 97         | 90         | 73         |             |

absolute null loss. Table XVII lists the CatBoost median by threshold and the informative counts. The middle row is the 0.01 rule behind the retention tables. Thresholds 0, 0.001, and 0.05 keep that median and that count. At 0.1 the count is 96 and the median is 0.942. Fit-seed variance across seeds other than 11 is not measured.

## VII. DISCUSSION

CTGAN and TVAE do not match the CatBoost retention, and their successful size- $4n$  artifacts miss the cap. None of these comparisons emits MFS-v2, PTF-v1, or a release at the 10,240-byte tier. A small Wilcoxon  $p$ -value does not flip that null. The 8,192-step budget moves the CatBoost median by 0.034 and is not the displayed model. BeyondArena fits 5 further families inside the cap and does not show retention. The unmeasured publications and the null release scores are unchanged from Section I-0a. An equal tuning budget across generators is not measured.

## VIII. THREATS TO VALIDITY

*a) Internal.*: The fits use the coreset in Section IV, fit seed 11 only, and a training-derived validation cut of the official training rows. The scored retention panel is regression only. Other tasks are not a result of these tables. The supplement records the draw. The official test was not an input to training, selection, or the numbers in this paper. The informativeness rule drops a negligible real-versus-null gap. Thresholds 0, 0.001, and 0.05 keep that median and that count. At 0.1 the count is 96 and the median is 0.942. Incomplete groups are omitted, not zeroed. Chow-Liu and independent marginals are implementations written for this comparison. CTGAN and TVAE use the lineages that finished. Runs that stopped at the recorded time limit are omitted. They are not counted as failures of a finished fit. The validation cut also supplies native tuning signals. These are post-selection validation estimates, not independent official-test estimates. Lineage intervals hold the split and fitted artifact fixed; they omit fitting uncertainty. Related simulation lineages can violate the independence assumption behind both bootstrap and order-statistic intervals. The retained-subset results therefore remain exploratory.

*b) External.*: These fits do not establish behavior on the full tables used by the diffusion papers, on five fit seeds, or on the official test. Generalization past seed 1729 is not measured.

Forest-Flow’s byte comparison uses 6 lineages and the hash-checked fit files. BeyondArena is a separate fit check. It is not a retention panel, and it is not pooled with the density block.

*c) Construct.*: Retention is the reported utility. The release score in Section III-D stays null. The linear auditor does not separate DOPE from the Gaussian copula, and that result is not an equivalence. Shipping this artifact is not the same experiment as shipping the compact predictor itself. The panel does not contain that control, so it does not show that a recipient needs the synthetic rows rather than the predictor. Normalized-space RMSE is the loss the auditors stored. Original-unit error after inverting the training transform was not stored, and this paper does not report it. Table VII is not a membership-inference certification. Its unbalanced DCR share and DOMIAS AUC are empirical. Empirical resemblance is not differential privacy and is not HIPAA de-identification.

*d) Statistical.*: The rank tests are descriptive, not the pre-registered production superiority test. Density, neural, Forest-Flow, and BeyondArena stay unpooled. A small retention  $p$ -value does not flip the null scalar.

## IX. CONCLUSION

On 97 regression lineages and fit seed 11, generator `features12_steps2048` has CatBoost retention 0.940 (interval 0.914 to 0.976) and 98/100 artifacts inside the cap. CatBoost separates it from the density comparators. The linear auditor does not separate it from the Gaussian copula, and that result is not an equivalence. The 6-lineage Forest-Flow block leads on CatBoost (1/0/5) and the MLP (0/0/4). Those counts are not a certified win. The unmeasured publications and the release scores are unchanged from Section I-0a. Official tests stay sealed.

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